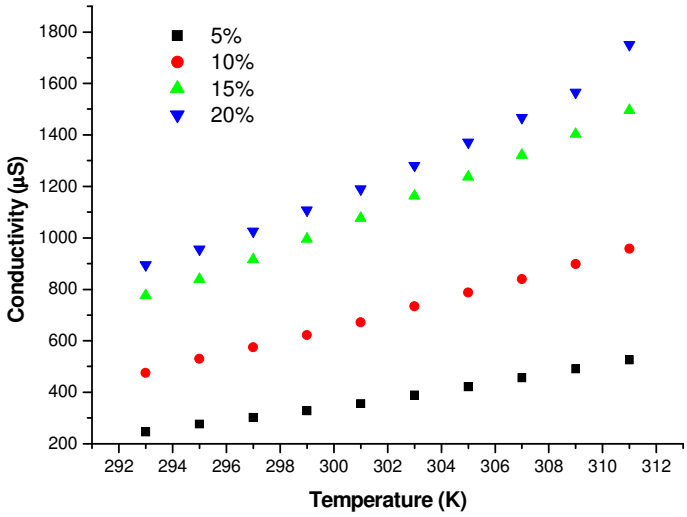


***Figure 7.16**: Plot of temperature versus conductivity as a function of choline chloride composition for the 2,3-butanediol system*

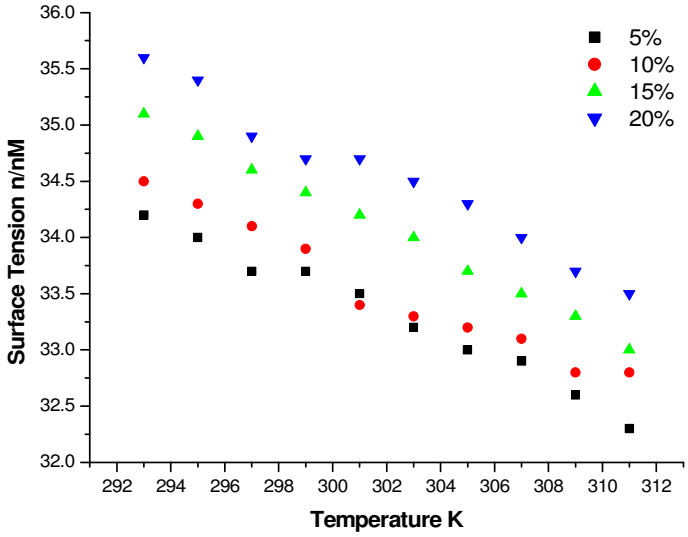


***Figure 7.17**: Plot of temperature versus surface tension as a function of choline chloride composition for the 2,3-butanediol system*

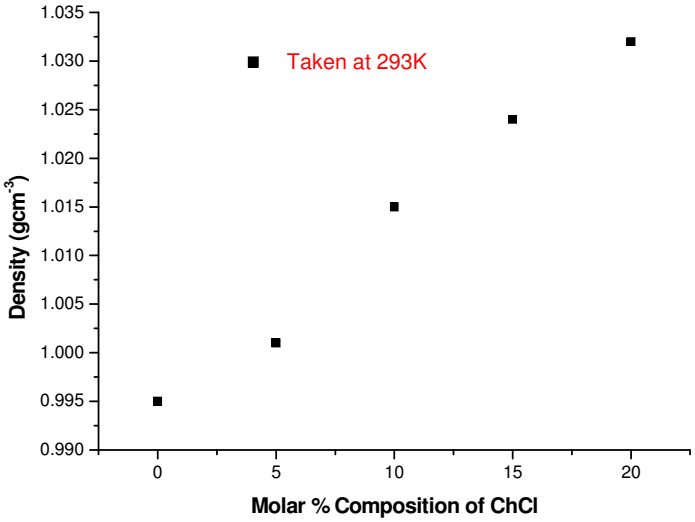


***Figure 7.18**: Plot of choline chloride composition versus density for the 2,3-butanediol system*

7.1.7 Choline Chloride : Glycerol

***Table 7.36:** Raw data for glycerol with 5% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	998	654	450	314	222	163	120	90
Conductivity (µS)	106.4	143.3	190.1	252	320	394	479	568
Surface Tension (n/nM)	63.7	62.8	62.0	61.3	60.5	59.8	59.3	58.7

***Table 7.37:** Raw data for glycerol with 10% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	790	503	349	247	180	132	98	74
Conductivity (µS)	243	297	374	483	601	759	980	1245
Surface Tension (n/nM)	61.6	61.0	60.3	59.5	59.0	58.4	57.9	57.5

Table 7.38: *Raw data for glycerol with 15% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	548	380	270	194	140	107	80	60
Conductivity (µS)	470	555	666	798	963	1114	1403	1788
Surface Tension (n/nM)	60.8	60.2	59.7	59.1	58.7	58.1	57.6	57.3

Table 7.39: *Raw data for glycerol with 20% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	503	350	246	178	132	98	86	57
Conductivity (µS)	580	739	930	1140	1362	1609	1923	2250
Surface Tension (n/nM)	57.4	57.0	56.6	56.1	55.7	55.2	54.8	54.4

Table 7.40: *Raw data for glycerol with 25% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	450	320	229	169	126	95	73	58
Conductivity (µS)	850	1030	1244	1490	1835	2140	2480	2930
Surface Tension (n/nM)	50.8	50.3	49.9	49.3	48.9	48.4	48.0	47.6

Table 7.41: *Raw data for glycerol with 30% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	401	295	213	160	120	91	71	55
Conductivity (µS)	964	1165	1410	1750	2080	2430	2800	3260
Surface Tension (n/nM)	48.5	48.0	47.5	47.1	46.7	46.4	46.0	45.6

Table 7.42: *Raw data for glycerol with 33% choline chloride*

Temperature (^oC)	**20**	**25**	**30**	**35**	**40**	**45**	**50**	**55**
Viscosity (cP)	376	259	188	140	104	82	64	52
Conductivity (µS)	1047	1298	1636	1947	2320	2710	3110	3560
Surface Tension (n/nM)	55.8	55.4	55.0	54.7	54.2	53.8	53.4	53.0

Table 7.43: *Raw data for density of the glycerol systems choline chloride at 293K*

ChCl %	**0**	**5**	**10**	**15**	**20**	**25**	**30**	**33**
Density (gcm^{-3})	1.261	1.242	1.234	1.219	1.210	1.203	1.192	1.181

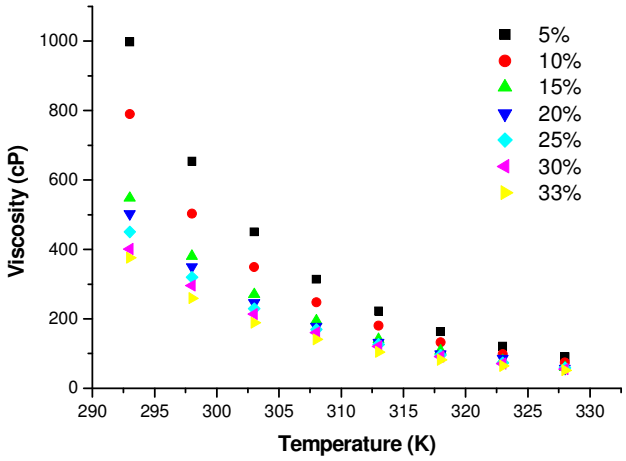


***Figure 7.19**: Plot of temperature versus viscosity as a function of choline chloride composition for the glycerol system*

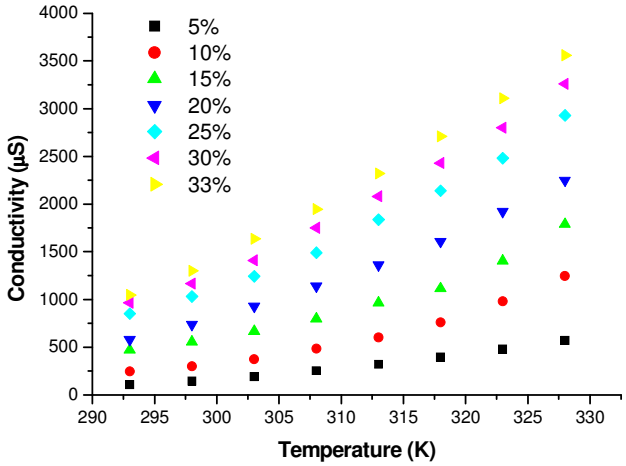


***Figure 7.20**: Plot of temperature versus viscosity as a function of choline chloride composition for the glycerol system*

7.2 Esterification Raw Data

7.2.1 Pure Glycerol

Table 7.44: *Raw data for concentrations of products and reagents in pure glycerol with 0.1 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.1	0	0	0
30	0.0777	0.0131	4.6E-4	0
60	0.08064	0.01644	4.3E-4	6.87E-4
90	0.07845	0.01923	3.05E-4	5.7E-4
180	0.07542	0.02132	5.44E-4	7.25E-4
240	0.07495	0.02203	6.72E-4	5.6E-4
300	0.07283	0.02428	8.53E-4	3.95E-4
360	0.07319	0.02322	0.00112	4.49E-4
420	0.07166	0.02389	0.00117	7.03E-4
480	0.07282	0.02445	8.31E-4	3.54E-4

Table 7.45: *Raw data for concentrations of products and reagents in pure glycerol with 0.2 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.2	0	0	0
30	0.08163	0.1027	0.00783	0
60	0.0317	0.12862	0.01913	2.58E-4
90	0.01876	0.13147	0.02421	4.54E-4
120	0.01063	0.1303	0.02885	4.56E-4
180	0.01446	0.11393	0.03457	8.25E-4
240	0.01967	0.11352	0.03163	0.00118
300	0.02151	0.11229	0.03117	0.00128
360	0.02217	0.1098	0.03209	0.00129
420	0.02457	0.10701	0.03166	0.0017
1440	0.02348	0.1099	0.03086	0.00163

Table 7.46: *Raw data for concentrations of products and reagents in pure glycerol with 0.3 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.3	0	0	0
30	0.11832	0.15416	0.01323	0
60	0.05133	0.17224	0.03764	3.8E-4
90	0.0274	0.19973	0.03552	6.09E-4
120	0.02016	0.19667	0.04021	9.16E-4
180	0.01635	0.19374	0.04314	0.00121
360	0.02841	0.16824	0.04838	0.0022
420	0.01995	0.17575	0.04909	0.00205
480	0.02369	0.16323	0.05291	0.00242
540	0.02944	0.15989	0.05161	0.00248
1620	0.02259	0.16888	0.04992	0.0029

Table 7.47: *Raw data for concentrations of products and reagents in pure glycerol with 0.4 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.4	0	0	0
30	0.2466	0.13435	0.00952	0
60	0.13384	0.21112	0.02698	3.6E-4
90	0.08704	0.20952	0.04983	0.00126
120	0.04818	0.24949	0.04855	0.00174
180	0.04839	0.22793	0.0577	0.00275
240	0.05138	0.21491	0.06153	0.00355
300	0.05047	0.21519	0.0626	0.00305
360	0.05221	0.22125	0.05743	0.0039
480	0.05299	0.22522	0.05554	0.00357
540	0.05928	0.21212	0.05875	0.00371
1560	0.04436	0.09188	0.0998	0.02139

Table 7.48: *Raw data for concentrations of products and reagents in pure glycerol with 0.6 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.6	0	0	0
30	0.34933	0.21333	0.01867	0
60	0.19603	0.31514	0.04367	0.00149
90	0.16588	0.21038	0.11025	0.00324
120	0.1439	0.23983	0.10356	0.00916
180	0.13291	0.25063	0.10158	0.01329
300	0.08773	0.24058	0.12859	0.01452
360	0.07565	0.24928	0.13001	0.01504
420	0.08421	0.27116	0.11362	0.0212
480	0.08642	0.27738	0.10989	0.01643
1320	0.06887	0.28838	0.11083	0.02109

7.2.2 ChCl : Glycerol 1:3

Table 7.49: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:3 with 0.2 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.2	0	0	0
30	0.1331	0.02163	0.02253	6.93E-5
60	0.11367	0.02002	0.03306	5.96E-5
120	0.04601	0.01456	0.06924	3.15E-4
180	0.04506	0.01245	0.0691	0.00143
300	0.02781	0.04424	0.06103	0.00197
420	0.02083	0.04427	0.06289	0.00304
480	0.02234	0.03951	0.06539	0.00245
540	0.01339	0.05209	0.0633	0.00264
1320	0.02074	0.02931	0.06381	0.00744

Table 7.50: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:3 with 0.3 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.3	0	0	0
30	0.25082	0.03344	0.00787	0
60	0.19644	0.05257	0.02527	0
300	0.03325	0.07539	0.08718	0.00535
360	0.03793	0.08118	0.08246	0.00751
420	0.03123	0.08443	0.08166	0.01041
480	0.02866	0.08225	0.08221	0.00846
540	0.03123	0.07894	0.08087	0.00987
1620	0.02808	0.06211	0.0797	0.00398

Table 7.51: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:3 with 0.4 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.4	0	0	0
30	0.36124	0.03173	0.00351	0
60	0.34009	0.04465	0.00763	0
90	0.26924	0.07708	0.02663	1.38E-4
120	0.20308	0.09983	0.04718	9.12E-4
180	0.10922	0.11996	0.08057	0.00322
240	0.0755	0.11552	0.0906	0.00926
300	0.08249	0.11516	0.09048	0.00713
360	0.07082	0.10865	0.09524	0.01002
420	0.05658	0.11159	0.09509	0.01388
480	0.05527	0.11975	0.09557	0.01128
540	0.05521	0.12128	0.09202	0.01316
1560	0.04494	0.19324	0.07296	0.0053

Table 7.52: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:3 with 0.6 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.6	0	0	0
30	0.41837	0.12392	0.02885	0
60	0.32682	0.11362	0.07769	0.00418
90	0.18465	0.15757	0.12113	0.01551
180	0.11359	0.16602	0.14592	0.02854
240	0.10677	0.11793	0.15394	0.06741
300	0.10547	0.0678	0.1839	0.05893
360	0.117	0.04945	0.17108	0.09139
480	0.10411	0.04219	0.17753	0.09863
540	0.09803	0.04448	0.17973	0.09803
1320	0.10363	0.05078	0.15855	0.12877

7.2.3 ChCl : Glycerol 1:2

Table 7.53: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:2 with 0.1 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.1	0	0	0
30	0.05366	0.03384	0.00608	1.15E-4
60	0.02809	0.04905	0.01092	3.4E-4
90	0.01789	0.04937	0.01537	6.73E-4
120	0.0124	0.04959	0.01768	8.84E-4
180	0.00726	0.05115	0.01881	0.00132
300	0.0046	0.0446	0.02207	0.00222
360	0.00367	0.04566	0.02186	0.00232
420	0.00363	0.04275	0.02306	0.0025
480	0.00363	0.03918	0.02465	0.00263
540	0.00369	0.04114	0.0227	0.00326
1320	0.00204	0.03409	0.0258	0.00409

Table 7.54: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:2 with 0.2 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.2	0	0	0
30	0.10859	0.04249	0.02437	6.25E-5
60	0.06714	0.06194	0.03475	4.73E-4
90	0.04468	0.06164	0.04578	7.02E-4
240	0.03707	0.0139	0.07272	0.00119
360	0.04504	0.01243	0.06803	0.00216
420	0.03053	0.01794	0.07237	0.00226
480	0.03813	0.01243	0.06966	0.00338
540	0.02454	0.02454	0.0689	0.00271
1320	0.02065	0.0254	0.06401	0.00865

Table 7.55: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:2 with 0.3 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.3	0	0	0
30	0.237	0.04362	0.00969	0
60	0.16051	0.08532	0.02657	3.4E-4
120	0.1297	0.03721	0.06506	9.92E-4
180	0.06917	0.05188	0.08417	0.00354
360	0.0466	0.04138	0.09399	0.00801
420	0.03552	0.04425	0.09605	0.00877
480	0.03938	0.05097	0.09162	0.0088
540	0.02973	0.02973	0.10092	0.0129
1500	0.0199	0.0489	0.08988	0.01714

Table 7.56: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:2 with 0.4 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.4	0	0	0
30	0.33232	0.02134	0.02317	0
60	0.27443	0.07717	0.02399	1.4E-4
90	0.23959	0.07197	0.04308	7.63E-4
180	0.11314	0.06171	0.10897	0.0024
240	0.07384	0.08528	0.11087	0.00638
300	0.07723	0.07377	0.11205	0.0083
360	0.06339	0.09649	0.10704	0.00868
420	0.06249	0.08808	0.10811	0.01107
480	0.0511	0.07971	0.11758	0.01137
540	0.03979	0.06822	0.12342	0.01501
1320	0.02507	0.07455	0.11588	0.02287

Table 7.57: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:2 with 0.6 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.6	0	0	0
30	0.57519	0.01969	0.00256	0
60	0.5548	0.02338	0.01091	0
90	0.44246	0.07632	0.04022	7.74E-4
120	0.34982	0.11307	0.06537	0.00636
180	0.23439	0.10794	0.11951	0.01864
240	0.16365	0.07944	0.15979	0.03733
300	0.14104	0.09176	0.16517	0.03687
360	0.13273	0.08296	0.16863	0.04706
480	0.11917	0.05164	0.18182	0.06555
540	0.11683	0.04221	0.18317	0.07462
1620	0.08547	0.00718	0.18444	0.13846

7.2.4 ChCl : Glycerol 1:1.5

Table 7.58: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1.5 with 0.1 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.1	0	0	0
30	0.05403	0.03127	0.00675	3.99E-4
60	0.03901	0.04122	0.00913	3.91E-4
90	0.02791	0.04605	0.01221	5.43E-4
120	0.02157	0.04754	0.01428	7.77E-4
180	0.01513	0.0478	0.01703	0.001
240	0.01113	0.0488	0.01772	0.00154
420	0.01519	0.04732	0.01678	0.00131
480	0.01097	0.04417	0.01955	0.00192
540	0.00705	0.04345	0.02133	0.00228
1320	0.00651	0.03491	0.02496	0.00316

Table 7.59: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1.5 with 0.2 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.2	0	0	0
30	0.19667	0	0.00167	0
60	0.18998	0	0.00501	0
90	0.17559	0	0.0122	0
120	0.1073	0.01003	0.0409	2.93E-4
180	0.0867	0.01164	0.04964	7.92E-4
360	0.07585	0.00212	0.05039	7.08E-4
420	0.0757	0.00172	0.05943	0.00124
480	0.07611	0.00238	0.05831	0.00163
540	0.06872	0.00248	0.06118	0.00215
1320	0.05465	0.00277	0.06766	0.00242

Table 7.60: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1.5 with 0.3 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.3	0	0	0
30	0.2297	0.02379	0.02306	1.27E-4
60	0.20146	0.04854	0.02477	1.42E-4
90	0.20625	0.04639	0.02345	1.51E-4
120	0.1961	0.02816	0.0373	3.79E-4
180	0.15292	0.02691	0.05695	0.00209
300	0.0911	0.02779	0.07781	0.00676
360	0.08007	0.03909	0.08192	0.00566
420	0.06677	0.04097	0.08474	0.00759
480	0.06198	0.04673	0.08397	0.00779
540	0.04851	0.04105	0.09145	0.00918
1560	0.02742	0.04135	0.08273	0.02193

Table 7.61: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1.5 with 0.4 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.4	0	0	0
30	0.37128	0.01034	0.00919	0
60	0.28968	0.05891	0.02561	6.43E-5
90	0.23405	0.07155	0.04616	6.92E-4
120	0.17208	0.08839	0.06648	0.00219
180	0.11478	0.09746	0.08609	0.0052
300	0.07199	0.10911	0.09775	0.0078
360	0.06067	0.0915	0.11119	0.00849
420	0.06042	0.09005	0.10645	0.01223
480	0.05998	0.08376	0.11091	0.01148
540	0.04762	0.07954	0.11497	0.0143
1320	0.0227	0.07252	0.11823	0.02277

Table 7.62: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1.5 with 0.6 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.6	0	0	0
30	0.59103	0.00245	0.00326	0
60	0.54251	0.00648	0.02551	0
90	0.45042	0.00538	0.07109	0.00217
120	0.35196	0.10226	0.07067	0.00443
180	0.28226	0.10341	0.09492	0.02448
300	0.19993	0.12899	0.11824	0.03461
360	0.19264	0.13634	0.1158	0.03939
420	0.18019	0.1094	0.13193	0.04655
540	0.16525	0.14338	0.12272	0.04593
1320	0.08326	0.06712	0.15705	0.13553

7.2.5 ChCl : Glycerol 1:1

Table 6.63: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1 with 0.1 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.1	0	0	0
30	0.08198	0.017	2.02E-4	2.02E-4
60	0.0816	0.01104	0.00123	0.00164
90	0.07479	0.02222	0.00105	2.92E-4
120	0.07211	0.0249	0.00101	3.21E-4
240	0.04492	0.03522	0.00153	0.0056
300	0.04636	0.03787	0.00243	0.00364
360	0.04159	0.04106	0.00262	0.00404
480	0.04336	0.04191	0.00321	0.00277
540	0.04173	0.04248	0.00376	0.00276
1320	0.03108	0.0427	0.00473	0.00559

Table 7.64: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1 with 0.2 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.2	0	0	0
30	0.1804	0.00115	0.00922	0
60	0.15172	0.00318	0.02255	0
90	0.12051	0.00809	0.03508	4.13E-4
120	0.11071	0.00842	0.03899	9.63E-4
180	0.07484	0.03857	0.03915	0.00276
240	0.06427	0.04658	0.04009	0.00299
300	0.05058	0.06266	0.03821	0.00345
360	0.0481	0.05962	0.03946	0.00445
420	0.04314	0.06171	0.04134	0.00415
480	0.03706	0.06622	0.04131	0.0047
540	0.03001	0.07213	0.04155	0.00492
1440	0.0159	0.04252	0.05367	0.01141

Table 7.65: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1 with 0.3 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.3	0	0	0
30	0.20344	0.0342	0.03069	3.25E-4
60	0.16131	0.06089	0.03796	6.33E-4
120	0.0995	0.08545	0.05393	0.00239
180	0.07517	0.12034	0.04689	0.00357
240	0.05322	0.10153	0.06575	0.00458
300	0.0509	0.09447	0.06829	0.00602
360	0.05316	0.1012	0.06425	0.00606
480	0.06161	0.06494	0.07418	0.00837
540	0.05514	0.06565	0.07582	0.00919
1320	0.02023	0.06205	0.08779	0.01405

Table 7.66: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1 with 0.4 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.4	0	0	0
30	0.38799	2.65E-4	0.00587	0
60	0.33362	0.02189	0.02224	0
90	0.26945	0.05422	0.03697	7.99E-4
180	0.17017	0.09265	0.06198	0.00441
240	0.12087	0.10744	0.07417	0.00778
300	0.12013	0.09901	0.07354	0.01126
420	0.09282	0.09409	0.08881	0.01183
480	0.0891	0.07563	0.09946	0.01212
540	0.08197	0.0831	0.09763	0.01323
1320	0.05714	0.0792	0.10226	0.01972

Table 7.67: *Raw data for concentrations of products and reagents in ChCl: glycerol 1:1 with 0.6 M lauric acid*

Time (min)	[Free Acid]	[Mono Ester]	[Di Ester]	[Tri Ester]
0	0.6	0	0	0
30	0.59354	0	0.00323	0
60	0.5431	0	0.02845	0
120	0.38201	0.03488	0.08969	0.00374
240	0.18514	0.11973	0.12971	0.0357
300	0.16968	0.09756	0.14493	0.04289
360	0.16869	0.02907	0.17388	0.0545
420	0.15867	0.05459	0.15464	0.07745
480	0.14941	0.05644	0.15534	0.08348
540	0.15876	0.05365	0.14562	0.09635
1320	0.08865	0.05096	0.15418	0.15203

7.3 Publications

Application of Hole Theory to Define Ionic Liquids by their Transport Properties, Andrew P. Abbott, Robert C. Harris, Karl S. Ryder, *J.Phys. Chem. B*, **2007**, *111*, 4910

4910 *J. Phys. Chem. B* **2007,** *111,* 4910−4913

Application of Hole Theory to Define Ionic Liquids by their Transport Properties†

Andrew P. Abbott,* Robert C. Harris, and Karl S. Ryder

Chemistry Department, University of Leicester, Leicester, LE1 7RH,United Kingdom

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Eutectic mixtures of quaternary ammonium salts with Lewis or Brønsted acids have been described as ionic liquid, but doubt exists over the compositional range for which this description is valid. In the current work, the conductivity, viscosity, density, and surface tension of a number of glycolic mixtures with choline chloride are measured over the mole fraction range 0 to 0.33. The data are fitted to hole theory, and it is proposed that the composition at which the measured conductivity matches the theoretical value is the point at which hole mobility becomes the dominant mechanism for charge mobility. For the mixtures of ethylene glycol and butanediol, this occurs at a ChCl mole fraction of approximately 0.2.

Introduction

Ionic liquids are without doubt in their hiatus at the beginning of the 21st century, and yet there is significant confusion about what an ionic liquid actually is. The recognized definition as an ionic material that is liquid below 100 °C leaves significant question as to what constitutes an ionic material. Some authors like to limit the definition to cations and so-called discrete anions, e.g., BF_4 and NO_3^-, although this is unsatisfactory, as it excludes the original research in the area on chloroaluminate systems. These are less easy to categorize as ionic, as they depend upon complex equilibria and consist of a variety of anionic species. The concentrations of the ionic species are strongly affected by the composition of the different components.

Others have classified ionic liquids into first and second generation ionic liquids,[1] where first generation liquids are those based on eutectics and second generation have discrete anions.[2] We have sought to further divide the first generation liquids into a variety of types depending on the nature of the Lewis or Brønsted acid that complexes the halide anion to form an ionic complex.[3] While there is some dispute whether eutectics with Brønsted acids constitute ionic liquids at all, there are others who seek to widen the description of ionic liquids to include materials such as salt hydrates.[4]

What is clear is that all ionic liquids form due to delocalisation of charge and this can be described by an equilibrium

$$\text{cation} + \text{anion} + \text{complexing agent} \leftrightarrow \text{cation} + \text{complex anion} \quad (1)$$

or potentially

$$\text{cation} + \text{anion} + \text{complexing agent} \leftrightarrow \text{complex cation} + \text{anion} \quad (2)$$

with the vast majority focusing on the former case.

The confusion arises from the magnitude of the equilibrium constant. For discrete anions such as BF_4^- and even $((CF_3SO_2)_2N)^-$, the equilibrium lies clearly to the right of eq 1. For eutectic based liquids, the equilibrium constant depends upon the strength of the Lewis or Brønsted acid such that a variety of complex anions are possible. Hence, all of the following could be described as ionic liquids

$$Cat^+Cl^- + AlCl_3 \leftrightarrow Cat^+ + AlCl_4^- \quad (3)$$

$$Cat^+Cl^- + \text{urea} \leftrightarrow Cat^+ + Cl^-\cdot\text{urea} \quad (4)$$

$$LiClO_4 + 3.5\ H_2O \leftrightarrow Li^+\cdot xH_2O + ClO_4^-\cdot yH_2O \quad (5)$$

but there must be a point where the Lewis or Brønsted acid is in considerable excess and the system becomes a solution of salt in the acid. A similar scenario also exists with the incorporation of diluents or impurities. Many ionic liquids with discrete anions are hydrophilic, and the absorption of water is found to sometimes have a significant effect upon the viscosity and conductivity of the liquids.[5−7] What is required is a simple method of distinguishing the properties of ionic liquids from those of a concentrated solution.

In the current work, we will present physical data on mixtures formed between quaternary ammonium halides and a variety of glycol hydrogen bond donors as a function of salt concentration. These will provide evidence for the change in properties from those of an ionic liquid to those of a solution.

Experimental

Choline chloride (ChCl) (Aldrich 99%) was recrystallized from absolute ethanol, filtered, and dried under vacuum. Ethylene glycol (EG), 1,4-butanediol (butylene glycol, BG), and glycerol (all Aldrich >99%) were used as received. The eutectic mixtures were formed by stirring the two components at 100 °C until a homogeneous colorless liquid was formed. The liquids were cooled at a rate of 1 °C min^{-1}, and the freezing point was taken as the temperature at which the first solid began to form. The viscosity was determined using a Brookfield DV-E viscometer fitted with a thermostated jacket. The conductivity and its temperature dependence were determined using a Jenway 4071 conductivity meter with temperature and conductivity probes. A Kruss K11 tensiometer equipped with a thermostatted jacket, was used for the surface tension experiments. The equilibrium geometry and volumes of the ions and hydrogen

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* Corresponding author. E-mail: Andrew.abbott@le.ac.uk. Fax: + 44 116 252 3789.

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APPENDIX

bond donors were calculated using a Hartree−Fock method utilizing an STO-3G model provided by commercially available software.[8]

Results

Angell[9] has shown that for ionic liquids with discrete anions the molar conductivity is inversely proportional to the viscosity of the liquid and this correlation is virtually independent of the cation and anion. This has been used as a method of defining the properties of an ionic liquid. We have recently used hole theory to show that Angell's observation is due to a different mechanism of charge transfer in ionic liquids from that encountered in ionic solutions or high-temperature molten salts.[10,11] Hole theory was initially developed by Fürth[12] as an alternate view of the liquid state and later expanded by Bockris et al. for use with molten salts.[13] The model was found to have serious limitations in molten salts because ionic association significantly affects ionic activity. A discussion of the various models for pure fused salts is given by Stillinger.[14] However, in ionic liquids the large ion size and low temperatures mean that hole sizes are comparatively small. The probability of ions being able to move is several orders of magnitude smaller than in molten salts and so the role of ionic association can be ignored for charge transfer. Under these conditions Hole theory is an excellent model for the mobility of charged species.

We have shown that under these conditions the viscosity of a fluid, η, can be modeled by assuming it behaves like an ideal gas,[10] but its motion was restricted by the availability of sites of the ions/molecules to move into. Hence, it was shown that

$$\eta = \frac{\mathrm{mc}/2.12\sigma}{P(r > R)} \tag{6}$$

where m is the molecular mass (for ionic fluids this was taken as the geometric mean), $\bar{c}$ is the average speed of the molecule ($=(8kT/\pi m)^{1/2}$), σ is the collision diameter of the molecule ($4\pi R^2$), and k is the Boltzmann constant. The probability of finding a hole of radius, r, greater than the radius of the solvent molecule, R, in a given liquid, ($P(r > R)$) is given by integration of eq 7[10]:

$$P\cdot \mathrm{d}r = 0.601a^{7/2}r^2e^{-\mathrm{ar}}\,\mathrm{d}r \tag{7}$$

where $a = 4\pi\gamma/kT$ and γ is the surface tension. The good correlation obtained between the calculated and measured viscosities showed that it is valid to think of the viscosity of ionic fluids as being limited by the availability of holes. It therefore follows that charge transfer should occur in the same manner, hence ion motion depends on the migration of holes in the opposite direction. Since the fraction of suitably sized holes in ambient temperature ionic liquids is very low the holes can be assumed to be at infinite dilution and can be described by combining the Stokes−Einstein and Nernst-Einsten equations (eq 8)[11]: where z is the charge on the ion and e is the electronic

$$\lambda_+ = z^2F\, e/6\pi\eta R_+ \tag{8}$$

charge. An expression can therefore be obtained for the molar conductivity of both ions: This explains Angell's observation

$$\Lambda = z^2F\, e/6\pi\eta(R_+ + R_-) \tag{9}$$

that all ionic liquids display a correlation between the molar conductivity and the fluidity.[9] Although the size of the ion is also a variable in eq 9, in practice all of the ionic species that form ionic liquids have relatively similar dimensions ($R \approx 3 - 4$ Å). Hole theory therefore potentially provides a method by which the dilution of ionic liquids can be quantified because eq 9 will no longer be valid when hole mobility does not limit conductivity.

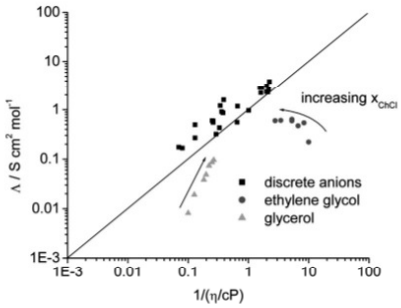


Figure 1. Plot of molar conductivity vs fluidity for a range of ionic liquids with discrete anions and mixtures of ChCl with glycerol and ethylene glycol at 20 °C.

Figure 1 shows a plot of molar conductivity versus fluidity ($1/\eta$) for the data presented in a recent review by Gali'nski et al. for ionic liquids with discrete anions.15 The scatter in Figure 1 is thought to be primarily due to the difference in size between the ions. It has previously been shown that eutectic mixtures of quaternary ammonium salts with hydrogen bond donors form liquids at ambient temperatures and the conductivity of these liquids obey eq 9.[16] We have assumed that these liquids can therefore be classified as ionic liquids and Figure 1 shows some that these data fit the classification of Angell.[9]

The ability to vary the composition of Lewis or Brønsted acid adds an additional dimension to the tuneability of the eutectic-based ionic liquids. To test the hypothesis that an ionic liquid can become a solution is relatively difficult, as most complexing agents are solid at ambient temperature and hence a true test should be conducted at the melting point of the complexing agent. This adds the additional complexity that the interaction between the anion and the complexing agent will be significantly affected by the temperature. Lecocq et al.[17] used ^{13}C and ^{35}Cl NMR spectroscopy to show that at 110 °C 1BMMICl:1$ZnCl_2$ varies its structure with time from $[BMMI]^+$-$[ZnCl_3]^-$ to [BMMI...Cl...$ZnCl_2$]. What is important, however, is the behavior of ionic liquids at ambient temperature. The only exceptions that we have found are with diols and triols which are liquid at ambient temperature and form relatively shallow eutectics when mixed with quaternary ammonium salts. These liquids have been shown to be very useful for electropolishing stainless steel as they are benign and simple to synthesize and recycle.[18,19]

Table 1 shows the conductivity, viscosity, density, and surface tension of three glycolic mixtures with ChCl: ethylene glycol, 1,4-butanediol, and glycerol. An interesting phenomenon occurs in the different liquids. For EG and BG, the viscosity increases as salt is added to the mixture as would be expected for most salt solutions. For glycerol, however, the viscosity decreases when ChCl is added. This is because the diols form linear aggregates of hydrogen-bonded molecules, whereas glycerol with its three OH moieties is a three dimensionally hydrogen-bonded liquid imparting greater order, higher density, and higher surface tension than the other glycols. The addition of salt to glycerol breaks up this structure by complexing the OH moieties to the anion of the salt. This causes a significant decrease in

APPENDIX

TABLE 1: Physical Properties of Various ChCl: Glycol Mixtures as a Function of Composition at 20.0 °C

ChCl: Ethylene Glycol							
% ChCl	5	10	15	20	25	30	33
density/gcm^{-3}	1.114	1.115	1.115	1.117	1.118	1.118	1.12
viscosity/cP	10	12	15	19	19	29	36
conductivity/mS cm^{-1}	3.74	7.52	8.14	8.53	8.74	7.92	7.61
surface tension/mN m^{-1}	48.1	47.3	45.3	47	45.4	47.2	49
ChCl: Glycerol							
density/gcm^{-3}	1.242	1.234	1.219	1.21	1.203	1.192	1.181
viscosity/cP	998	790	548	503	450	401	376
conductivity/μS cm^{-1}	106.4	243	470	580	850	964	1047
surface tension/mN m^{-1}	63.7	60.2	60.8	57.4	50.8	48.5	55.8
ChCl: 1,4Butanediol							
density/gcm^{-3}	1.021	1.026	1.036	1.046	1.052		
viscosity/cP	78	84	78	88	140		
conductivity/μS cm^{-1}	593	923	1271	1606	1654		
surface Tension/mN m^{-1}	46.4	46.8	46.9	47.4	47.6		

the surface tension opening up the free volume of the liquid, causing a decrease in the density and allowing greater movement of the ionic species.

Figure 2 shows the molar conductivity of choline chloride in a number of glycolic liquids. The molar conductivity was calculated by

$$\Lambda = \kappa M/\rho \tag{10}$$

where κ is the conductivity, ρ is the density, and M is the molar mass which is take as being the molar mass of the two components multiplied by their mole fraction in the mixture. The ionic radii were calculated using commercially available software assuming that each chloride ion is coordinated by one glycol molecule.[8] The only other viable assumption is that two glycol molecules coordinate the chloride ion. The difference in radii between the two complexes is approximately 0.6 Å, and this causes a change in the calculated molar conductivity of approximately 10%. FAB-MS shows the presence of the chloride coordinated to both one and two glycol molecules, and presumably the composition of the two species will depend upon the composition of the two components in the mixture.

At low concentrations, most of the ions exist as ion pairs, and the molar conductivity is relatively low. As the concentration of salt increases, the molar conductivity increases as the concentration of triple ion increases according to This is well-

$$3\mathrm{ChCl} \leftrightarrow \mathrm{Ch_2Cl^+} + \mathrm{ChCl_2^-}$$

known to occur in nonaqueous solutions.[20,21] As the salt concentration increases, the amount of free solvent decreases, and at some point there must be no free solvent molecules as

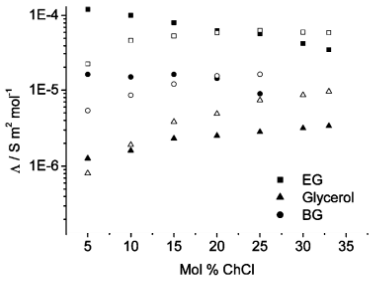


Figure 2. Plot of molar conductivity vs ChCl content for a variety of mixtures with different glycols at 20 °C. Open symbols are the measured values and filled symbols are the values calculated using eq 9.

they are all coordinated to the ions. At this point the movement of free "solvent" is prohibited, and it is only ionic species that can move. Under these conditions the mobility of the complex species is severely hindered due to their size and the decreased free volume of the liquid. For EG a maximum is observed in molar conductivity at approximately 25 mol % ChCl. According to eq 9, the limiting molar conductivity will be inversely proportional to the viscosity, and it would be expected that the molar conductivity would decrease as hole mobility replaced ion mobility as the dominant factor in charge transport. It is at this composition where hole mobility overtakes ion mobility as the dominant mechanism for charge-transfer that we propose an ionic liquid is formed.

Figure 1 shows the molar conductivity of the glycolic mixtures as a function of fluidity. It is clear that the molar conductivity values of the dilute solutions differ significantly from the Walden rule which is to be expected as the conductivity is dependent upon the number of charge carriers and the main mobile species which govern the fluidity of the liquid are the glycol molecules. Hence, the conductivity is greater than that predicted by eq 9. As the concentration of salt increases, the number of free solvent molecules decreases, and the conductivity approaches that of the other liquids where mobility is limited by hole mobility. Hence, the definition by Angell that all ionic liquids obey the same conductivity versus fluidity correlation is a good approximation to determining whether a material is an ionic liquid. A more rigid definition is possibly that an ionic liquid is a fluid where eq 9 is valid the justification being that in an ionic liquid the transport of charge is limited by hole mobility.

To determine the composition at which the eutectics in Figure 2 change from being ionic liquids to solutions of salt in glycols, the molar conductivity was calculated using eq 9 and plotted against the measured conductivity, and it can be seen that significant deviations are seen up to a composition of 20% EG and BG. Thereupon relatively good correlation was observed. BG could not be studied above 25 mol % because the temperature at which a homogeneous solution formed was above 20 °C. The other isomers of BG (1,2-, 1,3- and 2,3-) were also tested and showed similar trends to those observed for 1,4-BG, but these all reached supersaturation above 20 mol % at 20 °C and hence approached the properties of ionic liquids but could not exhibit the decrease in molar conductivity with concentration.

For glycerol the correlation between the values calculated from eq 9 and the measured values is closer a lower concentrations. This could be seen as an exception that shows the validity

APPENDIX

of the assumption, insofar as the very high viscosity of the pure fluid and the low free volume, as indicated by the high surface tension, mean that charge transport is still limited by hole transport, only in this exceptional case the mobility of the "solvent" molecules (i.e., glycerol) is not due to them all being complexed to the Cl^- but rather in this case they are hindered from moving by other solvent molecules.

If the mobility of charge is limited by the availability of suitably sized holes, then one diagnostic test for determining where most ionic liquids take on the properties of solutions would be that the molar conductivity passes through a maximum where ion mobility and hole mobility are at equilibrium.

Conclusions

It has been shown that mixtures of quaternary ammonium salts with glycols form homogeneous fluids with the properties of ionic liquids when the mole fraction of the salt exceeds 0.2. At higher salt concentrations, the dominant mode of charge transport is via the mobility of holes, and the mixture has the viscosity and conductivity characteristic of an ionic liquid, and below this concentration the properties can be described by the theories related to concentrated ionic solutions.

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Extraction of glycerol from biodiesel into a eutectic based ionic liquid, Andrew P. Abbott, Paul M. Cullis, Manda J. Gibson, Robert C. Harris, Emma Raven, *Green Chem.*, **2007**, *9*, 868

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Extraction of glycerol from biodiesel into a eutectic based ionic liquid

Andrew P. Abbott,* Paul M. Cullis, Manda J. Gibson, Robert C. Harris and Emma Raven

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A Lewis basic mixture of quaternary ammonium salts with glycerol has been used to extract excess glycerol from biodiesel formed from the reaction of triglycerides with ethanol in the presence of KOH. The effect of the cation on the partition coefficient of glycerol was determined, together with the time taken for the systems to reach equilibrium. Protocols were also investigated for the separation of glycerol from the quaternary ammonium salt mixtures.

Introduction

Biodiesel can be manufactured from a variety of natural product oil feedstocks, for example, sunflower oil, rapeseed oil, rubber seeds, and soy bean oil.[1–8] Its synthesis from a natural product means biodiesel is a carbon-neutral fuel (*i.e.* carbon dioxide generated from its combustion is equal to or less than the carbon dioxide used up by the plants it is derived from) and it produces significantly fewer particulates[1] than conventional diesel fuel with no sulfurous emissions. It can be used pure or blended with mineral diesel in most modern compression-ignition engines.[9]

The manufacturing procedure for biodiesel is much the same, regardless of the feedstock used. The triglyceride oil extracted from the plant is transesterified into alkyl esters (usually methyl or ethyl) using a catalyst such as KOH, NaOH, or a lipase. This yields three moles of ester and one mole of glycerol per mole of triglyceride as shown schematically in Fig. 1.[7]

The liberated fatty acid esters comprise a mixture of C-18 products, such as stearic, linoleic, linolenic, and oleic acid, and C-16 derivatives, such as palmitic acid in varying compositions depending on the source. The glycerol is an unwanted by-product and must be removed before the ester biodiesel can be used as fuel, as the viscosity of the glycerol present in the mixture impedes the high-pressure injection system of a modern diesel engine and may cause damage.

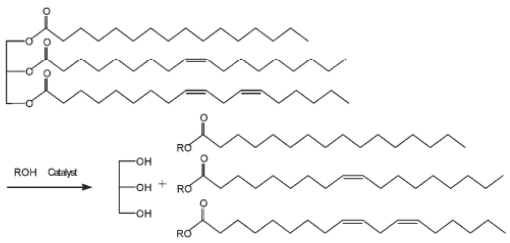


Fig. 1 Transesterification of triglycerides to alkyl esters of fatty acids and glycerol.

Chemistry Department, University of Leicester, Leicester, UK, LE1 7RH. E-mail: Andrew.abbott@le.ac.uk; Fax: +44 116 252 3789

Several methods have been used to remove glycerol from biodiesel and these include adsorption over silica,[10] membrane reactors,[11] and the addition of lime and phosphoric acid.[12] There are numerous problems associated with the costs and complications of operating biodiesel synthesis on an industrial scale.[7] Currently it is cheaper to drill and refine mineral diesel than to grow, extract, transesterify, and purify biodiesel, and methods to improve the economical feasibility of biodiesel production are sought.

The aim of this study is to design a system to separate glycerol and residual ethanol from some ethanol-esterified natural oils. We have previously shown that ionic liquids formed from eutectic mixtures of quaternary ammonium salts and hydrogen bond donors (HBDs) are an effective method of producing inexpensive, non-toxic, and environmentally benign solvent systems.[13,14] These so-called deep eutectic solvents (DESs) are made from quaternary ammonium salts mixed with a small HBD molecule, of which glycerol is one example. These DESs have been shown to have practical use in applications such as electropolishing[15] and metal oxide processing.[16]

The eutectic point (the composition of lowest freezing point and viscosity) of these solvents, that have so far been characterised, tends to occur at either 1 : 1 or 2 : 1 HBD : salt molar ratio.[13,14] The components of the mixture have a high affinity for each other due to the strength of the hydrogen bonding interactions formed. The high affinity of the salt for glycerol should mean that if a Lewis basic mixture of salt and glycerol is mixed with a phase with a low affinity for glycerol, then the former should be able to extract the excess glycerol. The high affinity of the ionic liquids for alcohols should also allow excess ethanol to be extracted from the biodiesel layer.

The aim of this work is to characterise and optimise a eutectic extraction system capable of washing glycerol out of the biodiesel product. Five quaternary ammonium salts were studied and all formed eutectics with glycerol that were miscible with water and immiscible with oils.

Experimental

The alcoholysis of vegetable oils was studied using alkaline catalysts. Potassium hydroxide, 5 g (0.9 mol) was suspended in 1 L of absolute ethanol and shaken until dissolved. 1 g of this solution was mixed with 5 g of oil and shaken in a small

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stoppered flask. The flasks were placed in an incubator shaker at 40 °C and 400 rpm for 24 h. After this time, they were allowed to stand for 1 h. Separation of the KOH-catalysed reaction in ethanol did not occur, but separation of the NaOH-catalysed reaction in methanol was observed. The molecular mass of soy bean oil was assumed to be 965.

A larger volume (10 ml oil and 2 ml 0.9 M KOH–ethanol) was produced under the same conditions and this was divided into 1 ml samples and used to test the eutectics and quaternary ammonium salts as extraction media. 1 mole equivalent of each salt was added to the oil samples and heated with agitation, but a lower layer was not seen to form. 1 : 1 eutectics of each quaternary ammonium salt were prepared and added to the solutions in a 1 : 1 salt : oil molar ratio.

For the larger scale experiments 500 g of oil (rapeseed or soy bean) and 150 g of 0.9 mol KOH in absolute ethanol were mixed in an incubator at 40 °C and 400 rpm for 24 h. After this period, a lower layer separated from the rapeseed oil reaction. This was collected and NMR analysis confirmed it to be predominantly glycerol. The soy bean oil reaction did not separate on standing.

The DES used for extraction was made by heating 46 g (0.5 mol) glycerol and 69.8 g (0.5 mol) of choline chloride on a hotplate at 50 °C until a homogenous transparent liquid formed. This was added to the homogeneous oil resulting from the soy bean oil reaction and the separated oil from the rapeseed oil reaction, shaken and allowed to stand for 30 min. Samples of the upper and lower layers were analysed by ^{1}H NMR in 1-butanol and dichloromethane, respectively, and a 10 μl sample of the upper layer was analysed by GC.

Removal of glycerol from a standard solution of ethyl caprate–glycerol was carried out using a standard solution of ethyl caprate–glycerol in a 3 : 1 molar ratio. To this 0.5% v/v solution, ethanol was added to aid miscibility. 10 ml of this solution was mixed with a 1 : 1 glycerol–acetylcholine chloride eutectic and samples were taken from the upper layer and analysed by FT-IR.

Viscosity and freezing point determination of deep eutectic solvents: viscosity was determined using a Brookfield DV-E viscometer according to the standard procedure. The freezing point was taken to be the glass transition temperature or crystallisation temperature as was measured with a Jenway 4071 temperature probe after gradual cooling in ice followed by dry ice–acetone.

The chloride ion concentration was determined using potentiometry with a silver chloride and calomel electrode.

In the cell

$$Hg|Hg_2Cl_2|KCl_{(sat)}||Cl^-{}_{analyte}\ Cl^-|AgCl|Ag$$

the change in potential with chloride ion concentration was found to follow a Nernstian behaviour.

Results and discussion

Soy bean oil was chosen as the feedstock of study because of its favourable cetane number, and also because it did not separate readily from glycerol at the end of the reaction, and therefore was useful for the principles to be studied. Fig. 2 shows a schematic of a process that uses a choline chloride–glycerol mixture to draw out excess glycerol from biodiesel.

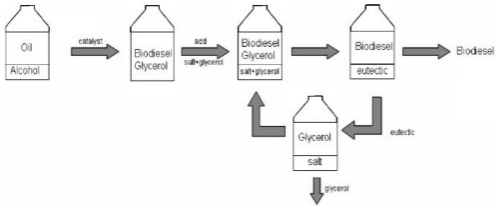


Fig. 2 Schematic diagram for the production of biodiesel and removal of excess glycerol using a sub-eutectic salt : glycerol mixture.

It was decided to concentrate on KOH and NaOH as the catalysts as these gave quantitative conversion of the oil to the methyl esters. Enzyme catalysed transesterification was also attempted, but of the fifteen lipases studied only *Candida antarctica* B gave a reasonable yield (38%) after a 24 h period of incubation. These data are comparable to the yields achieved with different oils published by Abigor *et al.*[17]

^{1}H NMR studies of the reaction products were undertaken using the solvent 1-butanol-d_{10}, as this was the only solvent in which glycerol, fatty acid esters, and unreacted soy bean oil were found to dissolve. The spectrum of the potassium hydroxide-catalysed reaction contained the glycerol signal but no discernible glyceride peaks (δ 4.15 and δ 4.3 ppm), which confirmed that the reaction had gone to completion.

Extraction of glycerol using sub-eutectic salt mixtures

Initially, pure quaternary ammonium salts were added to the crude biodiesel, to see if a deep eutectic would spontaneously form *in situ* extracting the glycerol and forming a separate layer. After the reaction was complete, the resulting oil was separated into 1 ml samples and each of these was shaken with a 0.5 mole equivalent of each of the quaternary ammonium salts chosen for the study. Upon shaking and heating, the salts all remained as solids and did not dissolve in the liquid phase, form a separate liquid phase, or effect any noticeable extraction of the glycerol from the biodiesel–glycerol mixture. This is probably due to the enthalpy of formation of the eutectic mixtures, and so an alternative approach was attempted.

Lewis basic mixtures of salt and glycerol (*i.e.* a 1 : 1 molar ratio) were made and heated until they formed homogeneous liquids. These were added to 1 ml samples of biodiesel in such quantity to ensure that the sum contents of both phases was a 1 : 2 salt : glycerol molar ratio. The mixtures were shaken, allowed to separate, and then analysed by ^{1}H NMR against the unwashed biodiesel product. A decrease in the glycerol signals (δ 3.55 and δ 3.45 ppm) in the oil phase was observed in all cases. No significant decrease in glycerol signals was observed when 1 : 2 salt : glycerol eutectics were added to the crude biodiesel in the same way, showing that the mixture is only effective at extracting glycerol when it is Lewis basic. No signal was detected for either ethanol or catalyst left in the biodiesel phase.

These initial results show that mixtures of salt and glycerol below the eutectic point can remove glycerol from crude

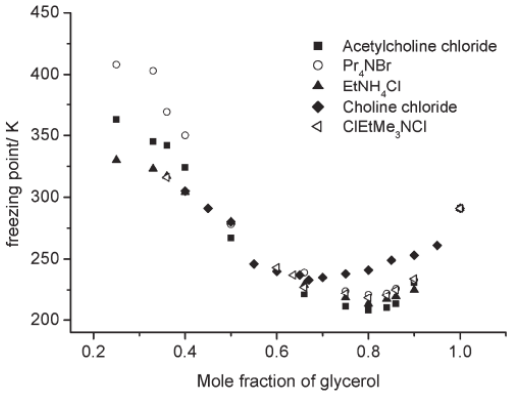


Fig. 3 Phase diagram of eutectics at different glycerol mole fractions.

biodiesel yields and suggests this extraction method to be suitable for more in-depth study.

Comparison of the physical properties of glycerol mixtures

The phase behaviour and viscosity of the salt : glycerol mixtures were characterised to identify which salts could be practically used to extract glycerol from biodiesel. The mixtures should preferably be liquid at ambient temperature and have viscosities that permit physical separation of phases by decantation or filtration. Five quaternary ammonium salts were used, and viscosities and freezing points of the mixtures were measured as a function of glycerol mole fractions in the range 0.35 to 1. The results obtained are shown in Fig. 3 and Fig. 4.

All of the eutectics studied showed their lowest viscosity at the mole fraction of approximately 0.75, whereas the freezing point minima tended to vary slightly depending on the salt. Choline chloride had a minimum freezing point at approximately 0.7, while the other salts had minima at mole fractions between 0.8 to 0.85. The freezing points at the eutectic point occurred between 210 and 235 K, meaning that all systems could be used effectively at ambient temperatures.

All the salts were also viable with respect to their viscosity. Since mixtures at the eutectic point were not found to be effective at removing glycerol from biodiesel, a Lewis basic mixture was needed. The 1 : 1 ratio of salt : glycerol mixture appeared to be the best compromise between viscosity and melting point and low mole fraction of glycerol. In this way, it is hoped that a 1 : 1 mixture of salt : glycerol could be used to extract one mole of glycerol from an equivalent 3 moles of biodiesel.

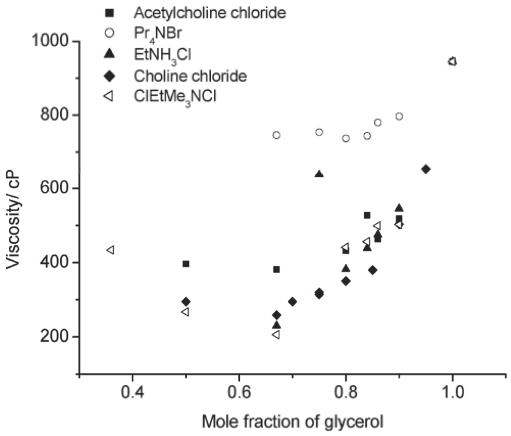


Fig. 4 Viscosity graph for eutectics at different glycerol mole fractions.

Quantification of glycerol extraction

The acetylcholine chloride eutectic was selected to study the extraction of glycerol from biodiesel as a function of time because the acetyl group on the cation and the OH group in glycerol have strong FT-IR resonances at 3330 and 1740–1730 cm^{-1}, respectively. These absorbances can be monitored over time to ascertain the time taken to reach equilibrium. By dividing the absolute intensity of the OH resonance by the absolute intensity of the acetyl resonance for studies of various molar ratios of the eutectic, a calibration curve was produced. A molar equivalent of 1 acetylcholine chloride eutectic to 1 glycerol was then used to wash a 1 ml sample of crude biodiesel. At various times, 10 μl samples of the eutectic layer were taken and analysed by FT-IR. The results are shown in Fig. 5.

This experiment revealed that the eutectic reached glycerol saturation after 10 min. Saturation corresponded to an approximate 1 : 2 molar ratio of acetylcholine chloride : glycerol and >99% removal of glycerol from the ester layer.

Effect of salt and initial composition of washing liquid

A standard mixture of 3 moles of ethyl caprate and 1 mole glycerol, with 5% v/v ethanol to aid miscibility, was mixed with

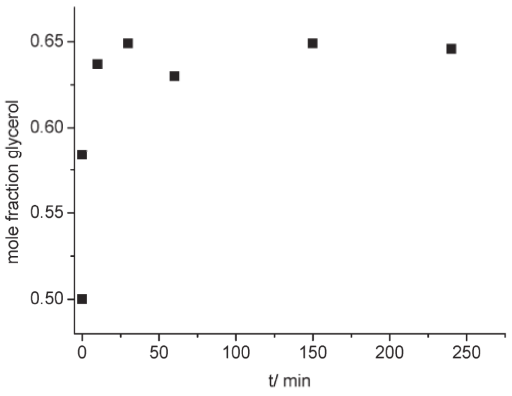


Fig. 5 Glycerol mole fraction in an acetylcholine chloride–glycerol mixture in contact with biodiesel containing glycerol as a function of time.

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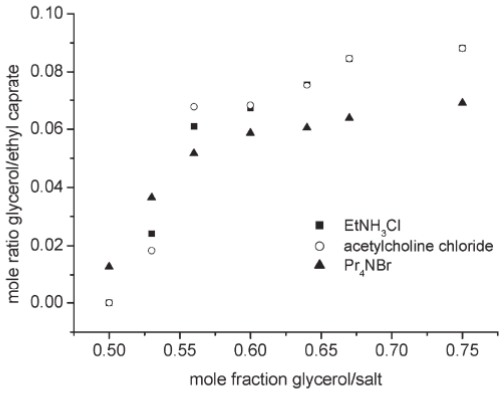


Fig. 6 Mole ratio of glycerol : ethyl caprate remaining in the biodiesel phase after washing with different salt–glycerol mixtures.

a variety of quaternary ammonium salt–glycerol mixtures of different molar ratios to determine their efficacy for glycerol removal. After shaking and warming the vials and allowing them to settle into two layers, the upper (ethyl caprate) layer was assayed by GC. The resulting glycerol–ethyl caprate mole fractions are shown in Fig. 6.

Fig. 6 shows similar profiles for the acetylcholine chloride and $EtNH_3Cl$ eutectics. Pr_4NBr shows a slightly different profile with a better extraction of glycerol at higher mole fractions but a weaker extraction efficacy at the lower (0.5) mole fraction. From these results, it would appear the ability of the salt to draw glycerol out of the other phase may be dependent on the anion but is unaffected by the cation. The ability of chloride to extract glycerol at the 0.5 mole fraction is likely due to the higher electronegativity of the ion compared to bromide. Conversely, the lower ionic radius of chloride may become more of an issue at higher mole fractions of glycerol, and thus bromide is more effective at extracting glycerol at these mole fractions because its larger ionic radius simply means more moles of glycerol will fit around it spatially.

These results showed that the ability of the eutectics to remove glycerol from a standard system decreased with increasing glycerol concentration. The 1 : 1 molar ratio of glycerol : salt was found to be the most suitable in terms of its extraction ability and its ease of use. The three salts all gave very similar partition coefficients.

Extraction of glycerol from biodiesel into a eutectic solvent

The ideas developed with the ethyl caprate standard were applied to a practical sample of biodiesel produced using KOH, ethanol and soy bean oil. Mixtures of glycerol and salts were prepared in 1 : 1 molar ratios and used to extract 1 ml samples of crude biodiesel. The biodiesel layer was then analysed by GC to calculate the mole fraction of glycerol remaining. The results are shown in Table 1.

A significant decrease was observed in the mole fraction of glycerol remaining in the biodiesel layer after washing with all of the salt–glycerol mixtures. Particularly effective removal

Table 1 Mole fraction of glycerol in the biodiesel layer after washing with 1 : 1 glycerol : salt mixtures

Washing eutectic	$X_{glycerol}$
No wash	0.15
Pr_4NBr	0.08
$EtNH_3Cl$	0.00
$ClEtMe_3NCl$	0.00
Choline chloride	0.05
Acetylcholine chloride	0.03

was noted with $EtNH_3Cl$ and $ClEtMe_3NCl$, with effectively complete removal of glycerol from the biodiesel. The least effective salt was Pr_4NBr, which confirms the earlier result using the standard ethyl caprate system in Fig. 6.

^1H NMR of the washed biodiesel and eutectic mixtures showed no traces of cross contamination of the two phases. However, the biodiesel yields showed impurities on the GC trace after washing with Pr_4NBr eutectic, thought to result from possible contamination with Pr_3N, and likewise after washing with acetylcholine chloride eutectics, the biodiesel layer showed evidence of impurities on the NMR spectrum, corresponding to the signals on the NMR spectra of the free amines of these salts which almost certainly comes from trace impurities in the original quaternary ammonium salts. No evidence of contamination, however, was seen when choline chloride, $ClEtMe_3NCl$, or $EtNH_3Cl$ were used in washing solvents. Taking all these factors into account, $ClEtMe_3NCl$ and $EtNH_3Cl$ appear to be the most suitable quaternary ammonium salts for making eutectics for the extraction of glycerol from biodiesel, as they both remove almost all of it to practical detection limits in this system.

Separation of salt from glycerol mixture

To make the separation of glycerol from biodiesel using this method a viable process, a method of recovering the glycerol and the quaternary ammonium salt from the washing eutectic is necessary. In the system that has been developed here, the glycerol is recovered as a 2 : 1 mixture with a salt. In order to recycle the salt, it must be recovered or reduced to at least a 1 : 1 molar composition and the rest of the glycerol separated.

Possible purification methods include recrystallisation of the choline chloride, either by cooling or by addition of an antisolvent, and distillation of the glycerol. Distillation would be energetically difficult due to the high boiling point of glycerol (182 °C at 20 mm Hg) although vacuum distillation may be ultimately the easiest solution.

On cooling in ice, 1 : 2 mixtures of choline chloride : glycerol and acetylcholine chloride : glycerol both formed needle-like crystals. Filtration and analysis by the electrochemical method, however, showed that only about 4 wt% of the quaternary ammonium chloride could be recovered with either system. Further cooling of the solution in dry ice–acetone mixtures produced a glass-like material with no crystallisation.

Gas expansion of the mixtures was attempted using CO_2 at 50 bar as an antisolvent. The gas was soluble in the liquids as evidenced by opalescence of the solution on pressurisation and vigorous degassing on depressurisation, but no precipitation or crystal formation was observed over 24 h under pressure,

Table 2 Yield of biodiesel resulting from the transesterification of 550 g samples of two vegetable oils together with the amount of glycerol remaining after washing with a 1 : 1 glycerol : choline chloride mixture

Oil	Yield of biodiesel/g	Glycerol (wt%)
Soy bean	465	0.06
Rapeseed	471	0.02

and the potentiometric analysis showed the solution to be unchanged.

An alternative approach was attempted by adding a liquid co-solvent to aid the separation of the salt on cooling. 1-Butanol was added to the 1 choline chloride : 2 glycerol mixture, warmed to 50 °C to aid mixing and then cooled in ice. Significantly, more choline chloride formed as a precipitate than just by cooling the eutectic mixture. Analysis of the filtrate showed that 25% of the salt could be recovered by this method. While still unoptimised, the process shows that an anti-solvent could be added to recover the quaternary ammonium salt.

Scaled-up biodiesel synthesis with eutectic solvent extraction

To assess the suitability using salt–glycerol mixtures to purify biodiesel, a larger scale batch procedure was carried out. Both rapeseed and soy bean oil (500 g) were used with 150 g of KOH and ethanol for 24 h. Table 2 shows the yield of biodiesel produced from each experiment together with the amount of glycerol remaining after extraction with a 1 mole equivalent of 1 : 1 choline chloride : glycerol eutectic. After separation, the upper, biodiesel layer was analysed by ^{1}H NMR and GC-MS.

This shows that glycerol can be effectively removed from a biodiesel product on a moderate scale using a eutectic mixture.

Conclusions

Quaternary ammonium salt–glycerol mixture solvents were found to be successful as extraction media for glycerol from biodiesel product mixtures. A 1 : 1 glycerol : salt molar ratio was found to be the most effective for extracting glycerol from the biodiesel product. Of the salts studied, choline chloride, $ClEtMe_3NCl$, and $EtNH_3Cl$ showed the best results in terms of effective glycerol removal. Choline chloride was recovered with some success from a 1 : 2 mixture with glycerol by recrystallisation from 1-butanol, although this process was not optimised.

The process was scaled-up to a 500 ml batch process, with full alcoholysis of the triglyceride starting material and glycerol was successfully extracted by a choline chloride–glycerol washing solvent. Overall, the use of quaternary ammonium salt–glycerol DESs to remove glycerol by-products from the finished biodiesel reaction was found to be successful, but it is hoped further research will reveal a better method for recycling the salt and recovering the glycerol afterwards.

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